

EX NO:2

Chebyshev IIR Filter

2.1. Design and Performance Evaluation of a Chebyshev Type-I IIR Low-Pass Filter using MATLAB Program (Matlab Code):

```
clc; clear;

Wp=0.4; Ws=0.55; Rp=1; As=40;

[N,Wn]=cheb1ord(Wp,Ws,Rp,As);

[b,a]=cheby1(N,Rp,Wn);

fprintf('Filter Order = %d\n',N);

fprintf('Cutoff Frequency = %.2f * pi rad/sample\n',Wn);

if all(abs(roots(a))<1)

disp('System is STABLE');

else

disp('System is UNSTABLE');

end

% ----- GRAPHS -----

figure; freqz(b,a);

    % Magnitude + Phase

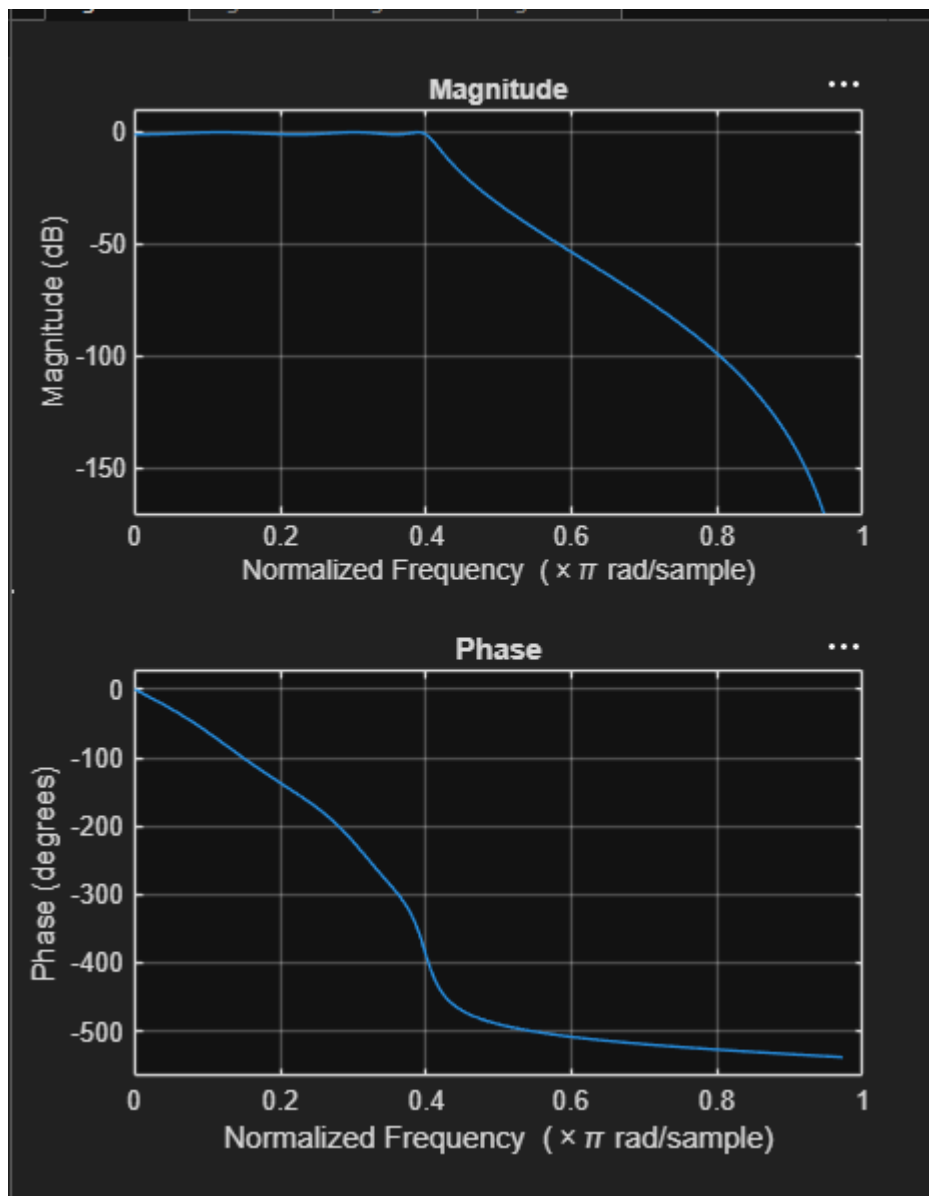
figure; impz(b,a);          % Impulse Response

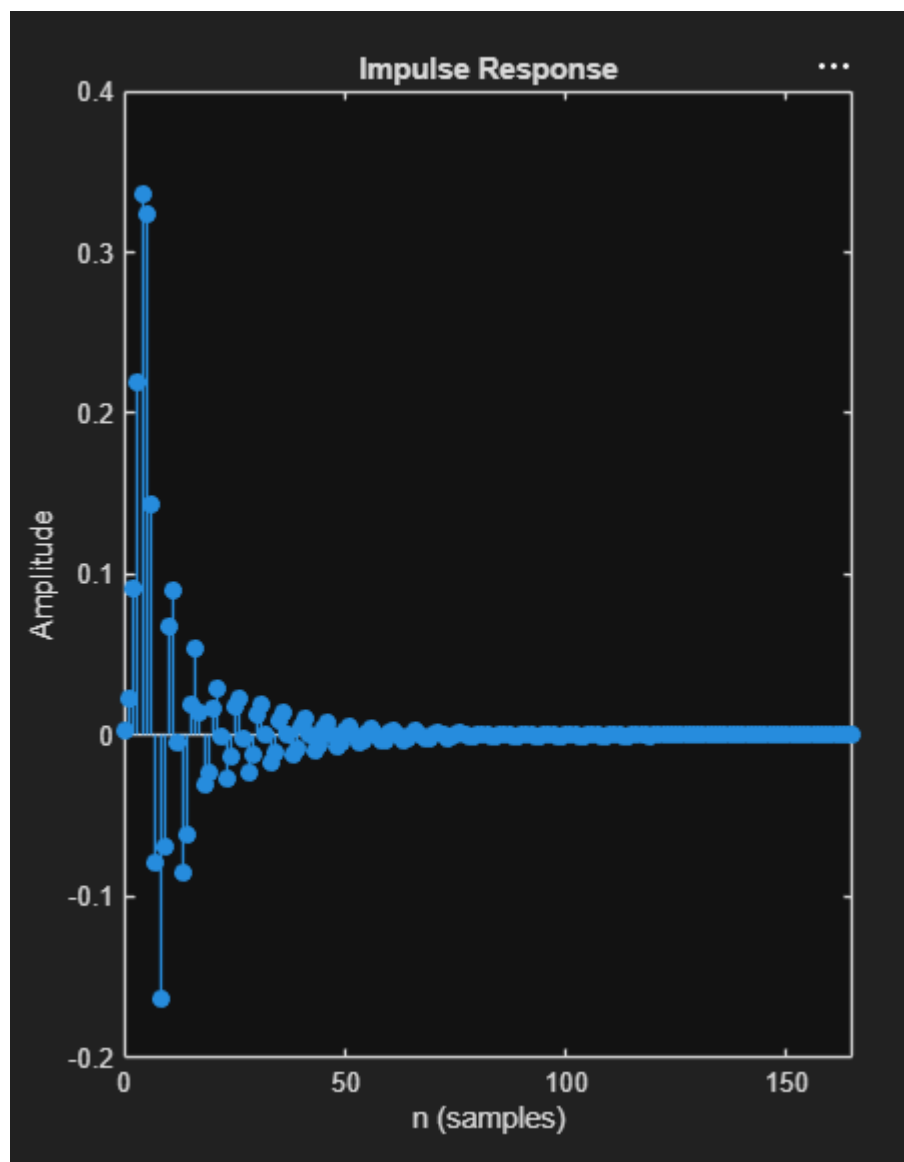
figure; grpdelay(b,a);

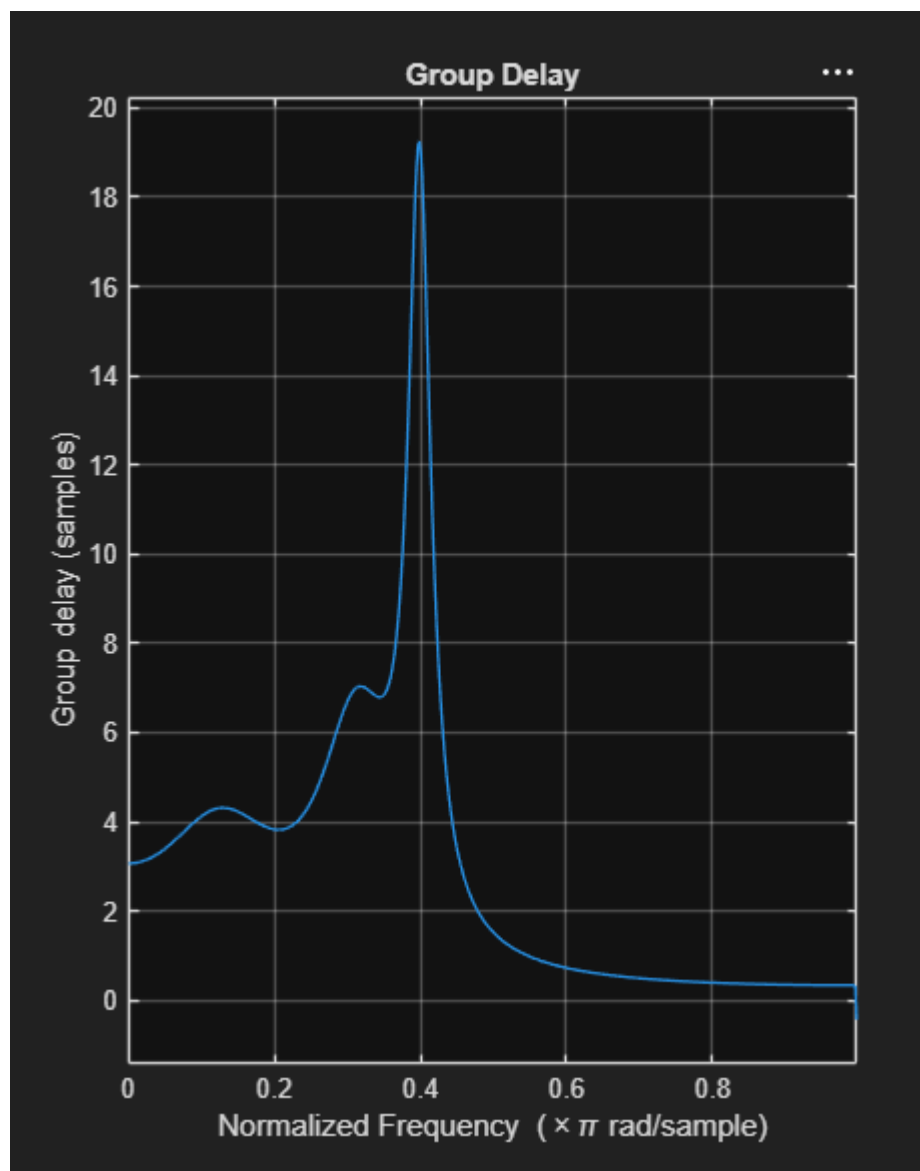
    % Group Delay

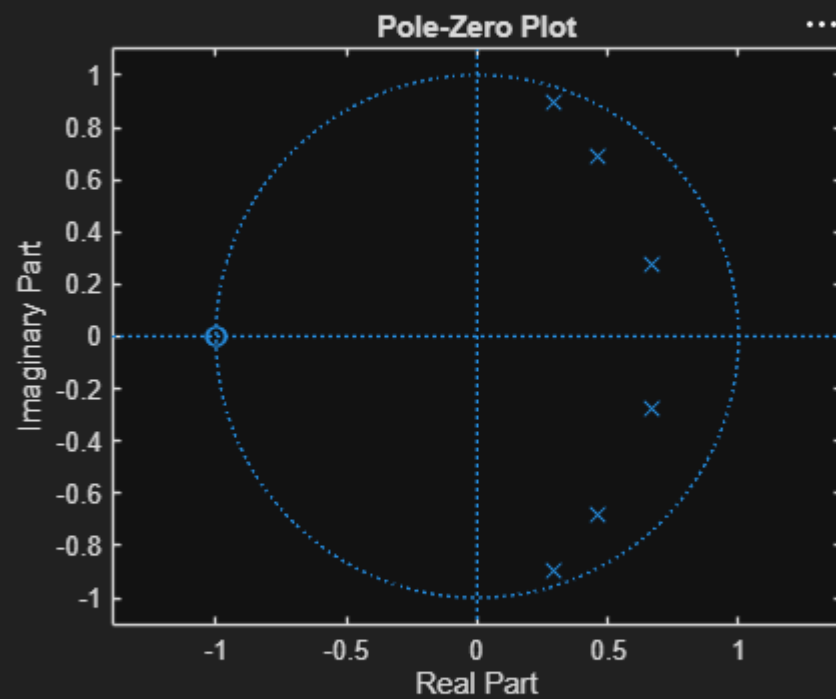
figure; zplane(b,a);        % Pole-Zero
```

Output:









Filter Order = 6
Cutoff Frequency = $0.40 * \pi$ rad/sample
System is STABLE

2.2. Design and Performance Evaluation of a Chebyshev Type-I IIR High-Pass Filter

Program (MATLAB Code)

```
clc;
clear;
close all;

% Specifications
Rp = 1;    % Passband Ripple (dB)
Rs = 40;   % Stopband Attenuation (dB)
Wp = 0.5;  % Passband Edge
Ws = 0.35; % Stopband Edge

% Find Filter Order
[n,Wn] = cheb1ord(Wp,Ws,Rp,Rs);

% Design High-Pass Filter
[b,a] = cheby1(n,Rp,Wn,'high');

% Frequency Response
figure;
freqz(b,a);
title('Frequency Response');

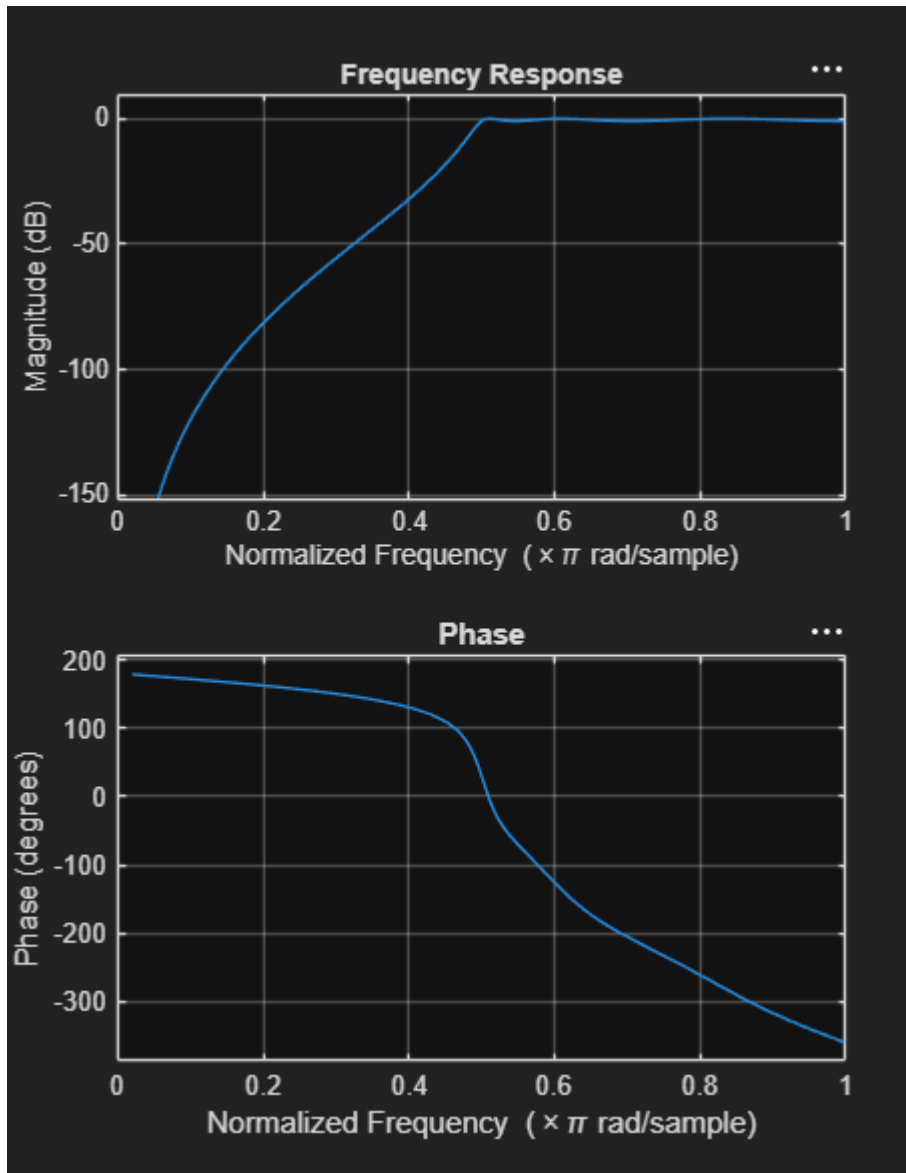
% Impulse Response
figure;
impz(b,a);
title('Impulse Response');

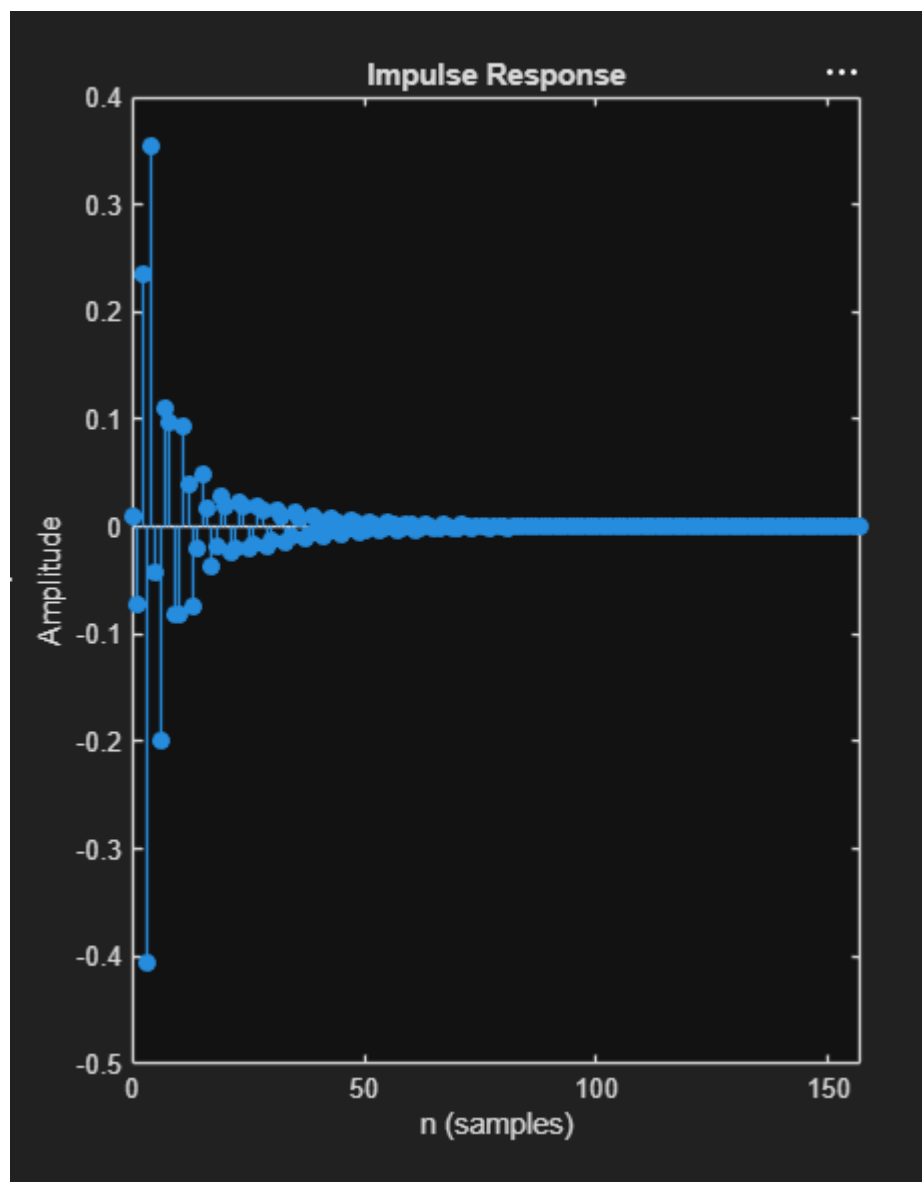
% Group Delay
figure;
grpdelay(b,a);
title('Group Delay');

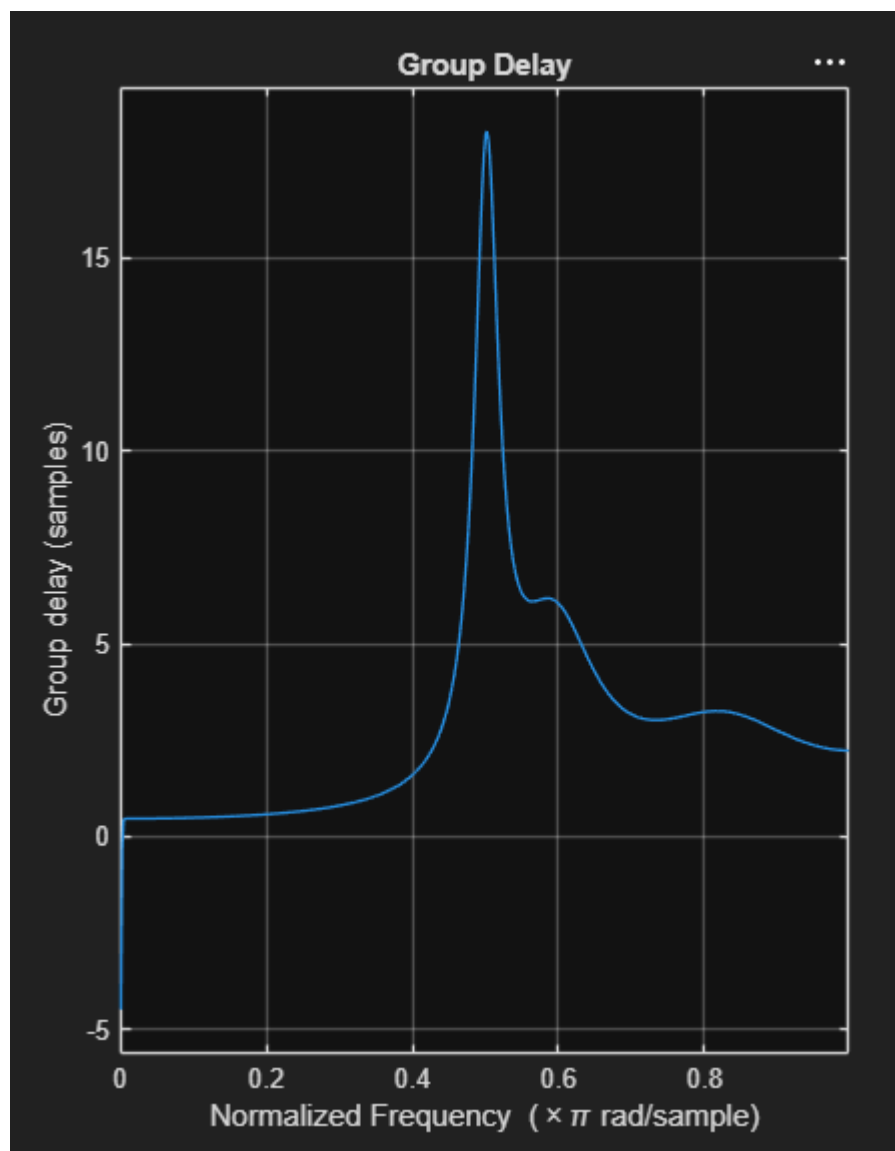
% Pole-Zero Plot
```

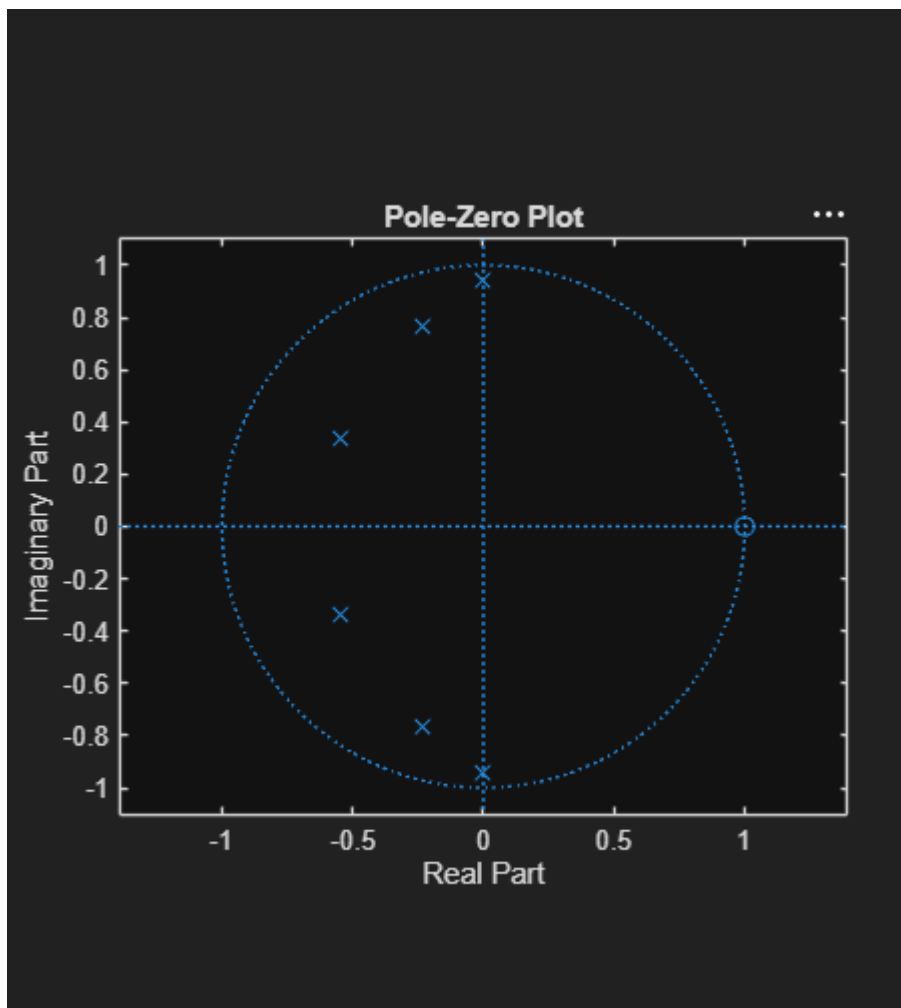
```
figure;  
zplane(b,a);  
title('Pole-Zero Plot');  
fprintf('Filter Order = %d\n',n);  
fprintf('Cutoff Frequency = %.4f\n',Wn);
```

Output:









Filter Order = 6
Cutoff Frequency = 0.5000
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